

Camp Fire Wind Model Comparison

Executive Summary

This analysis evaluated the performance of the baseline 3km High-Resolution Rapid Refresh (HRRR) forecast against a 1km dynamically downscaled wind field generated by WindNinja during the critical ignition phase of the 2018 Camp Fire (November 8, 2018). Both models were compared against ground truth data from Remote Automated Weather Stations (RAWS) at Jarbo Gap, Concow, and Paradise.

Key Findings

1. Underestimation of Peak Gusts:

- Both the 3km HRRR and the 1km WindNinja models significantly underestimated the extreme peak gusts observed during the event.
- The baseline HRRR 3km model underestimated peak gusts by an average of 23.05 mph.
- The WindNinja 1km downscaled model reduced this underestimation to 19.30 mph.

2. Model Improvement via Downscaling:

- Applying WindNinja's topographic downscaling provided an average error reduction of 3.75 mph, which translates to an overall 16.3% improvement in forecast accuracy over the baseline HRRR model.
- Improvement varied by location, with Concow seeing the most significant benefit (24.7% improvement), while Jarbo Gap and Paradise saw improvements of ~13.7% and ~13.8%, respectively.

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3. Statistical Significance:

- A paired t-test confirmed that the reduction in underestimation error provided by WindNinja is statistically significant ($p = 0.0004$), indicating that the downscaling process genuinely improves the representation of wind speeds in complex terrain, despite the persistent overall underprediction.

4. Impact of Elevation:

- The analysis revealed a strong negative correlation ($r \sim -0.92$) between station elevation and model underestimation. This suggests that the models struggle more severely at lower elevations or specific topographical features (e.g., canyons/slopes) compared to higher, more exposed ridges, though the small sample size warrants further investigation.

Conclusion

While standard meteorological models like the 3km HRRR capture broad synoptic trends, they inherently lack the resolution to resolve topographically-driven microclimate extremes, such as the severe gap winds experienced during the Camp Fire.

Dynamic downscaling tools like WindNinja offer a statistically significant, measurable improvement (averaging a 16.3% reduction in error) by coupling the broad wind flow with local terrain features. However, analysts must remain aware that even downscaled models can substantially underpredict extreme localized gusts. Ground truth observations and local terrain knowledge remain irreplaceable for accurate fire behavior prediction and situational awareness.